

EFFICIENT MEDICAL IMAGE COMPRESSION USING DISCRETE TRANSFORM

REVIEW – 02

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OVERVIEW



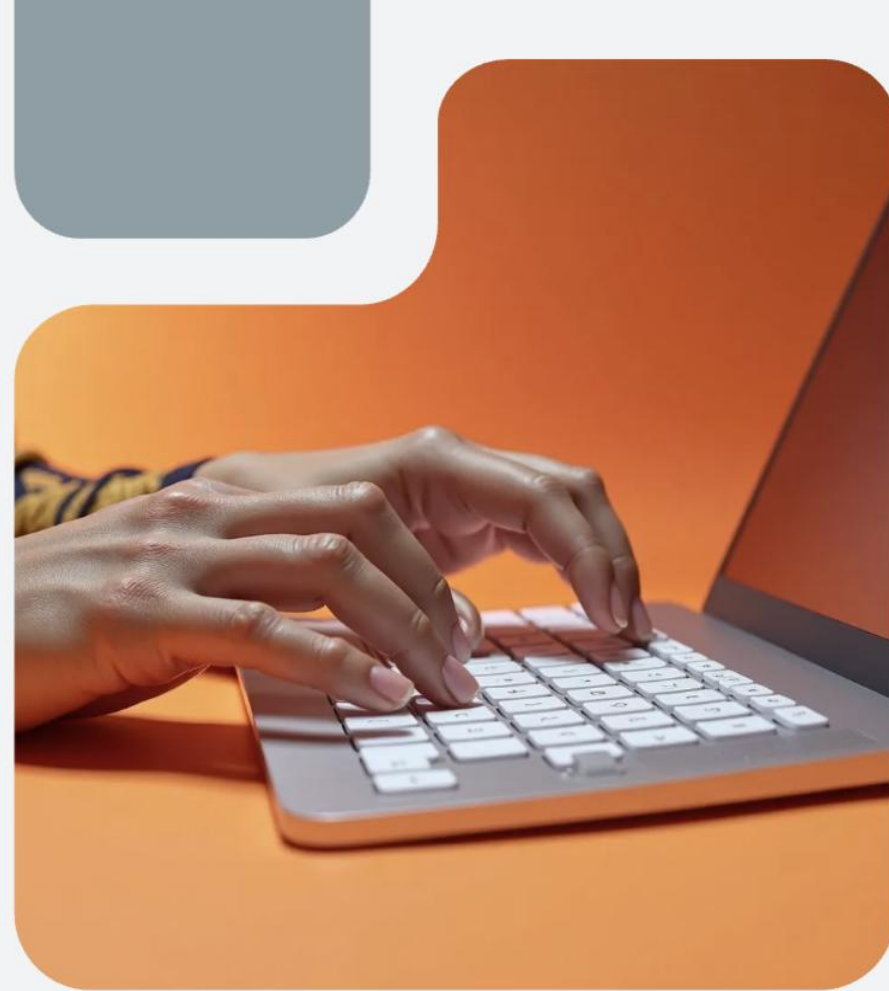
Literature Survey (10M)



30% Implementation (5M)



Module Preparation (5M)



Literature Survey



Ultrasound Image Compression Using Wavelet Coding (2025)

Goal: Improve the compression of ultrasonic images in noisy settings.

Methodology: Noise-resilient coding combined with wavelet-domain compression.

Ultrasound imaging is the dataset and modality. Important

Findings: Maintains diagnostic quality in loud environments.

Strengths: Real-time capability and resilient to noise.

Limitations: Non-ultrasound data results in decreased performance.

Relevance: Beneficial for portable diagnostic devices and tele-ultrasound



Progressive Medical Image Compression Using 2D-DWT and EZW (2025)

Goal: Use progressive image transmission to achieve effective compression.

Methodology: EZW (Embedded Zerotree Wavelet) in conjunction with 2D-DWT.

General medical imaging is the dataset and modality.

Important discoveries: Enables scalable decoding at various resolutions.

Strengths: Minimal bandwidth and storage needs. Cons: Not entirely tailored to 3D medical data. Relevance: Facilitates telemedicine and remote diagnosis



Comparative Analysis of Wavelet Bases for Medical Image Compression (2024)

Finding the best wavelet bases for compressing medical images is the goal.

Methodology: Wavelet families are compared. MRI and CT scans are the dataset and modality.

Important Results: The ideal ratio of compression to quality is attained by biorthogonal wavelets.

Strengths: Offers a methodical approach to wavelet selection.

Restrictions: Only applicable to 2D datasets. Relevance: Recommendations for designing adaptive wavelet-based compression



Hybrid DWT and VQ Technique for MRI/CT Compression (2019)

Goal: Increase MRI/CT scan compression efficiency.
Methodology: Vector quantization and DWT hybrid.
MRI and CT scans are the dataset and modality.
Key Findings: At high ratios, compression is almost lossless.
Strengths: Simple design and excellent performance.
Limitations: Strictly lossless applications are not appropriate.
Relevance: Good trade-off between diagnostic quality and storage efficiency



Wavelet Difference Reduction (WDR) Coding for Medical Images (2023)

Goal: Reduce compression artifacts while maintaining medical quality.

Methodology: Wavelet Difference Reduction (WDR) coding.

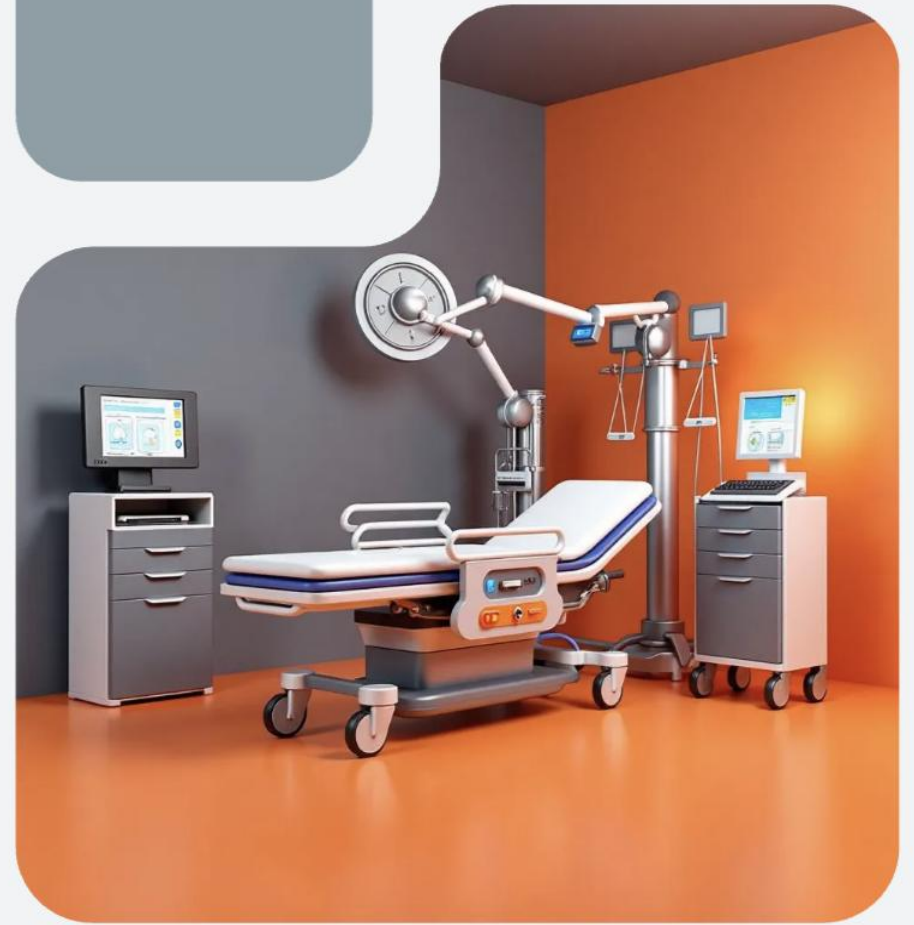
Dataset / Modality: Medical images across modalities.

Key Findings: Reduces redundancy and improves perceptual quality.

Strengths: Minimizes artifacts, better near-lossless results.

Limitations: Does not guarantee perfect lossless compression.

Relevance: Useful for clinical applications where near-lossless is acceptable



Deep Learning-Based Lossless Compression for CT/MRI (2025)

Goal: Use deep learning to deliver effective lossless compression.

Methodology: A coding scheme based on neural networks.

Dataset/Modality: Volumetric CT and MRI datasets.

Key Findings: More efficient than conventional wavelet techniques.

Strengths: Greater adaptability and complete losslessness.

Limitations: Needs a lot of computation and training data.

Relevance: Holds promise for medical imaging systems powered by AI



3D Predictive Coding for Volumetric Medical Image Compression (2017)

Goal: Providing lossless compression for volumetric medical photos.

Methodology: entropy coding combined with 3D predictive coding.

dataset and modality used: CT and MRI volumetric images

Important Results: robust lossless compression with redundancy between slices.

Strengths: Effectively decreases volumetric redundancy and is completely lossless. Limitations: Excessive memory utilization and prediction complexity.

Relevance: Volumetric compression systems' classical reference



Motion-Compensated Integer Wavelet Transform for 4D MRI (2005)

Goal: Make it possible for dynamic medical sequences to be compressed from lossy to lossless.

Methodology: Integer wavelet transform with motion compensation.

Modality/Dataset: 4D MRI scans.

Key Findings: Handles temporal correlations to achieve reversible compression.

Strengths: Good for medical imaging time series.

Limitations: Longer sequences require more computation.

Significance: The initial basis for dynamic compression of medical images



DWT-Based Lossless Compression Scheme for Radiology (2022)

Objective: Deliver high-precision lossless compression for medical imaging.

Methodology: DWT-based lossless compression scheme.

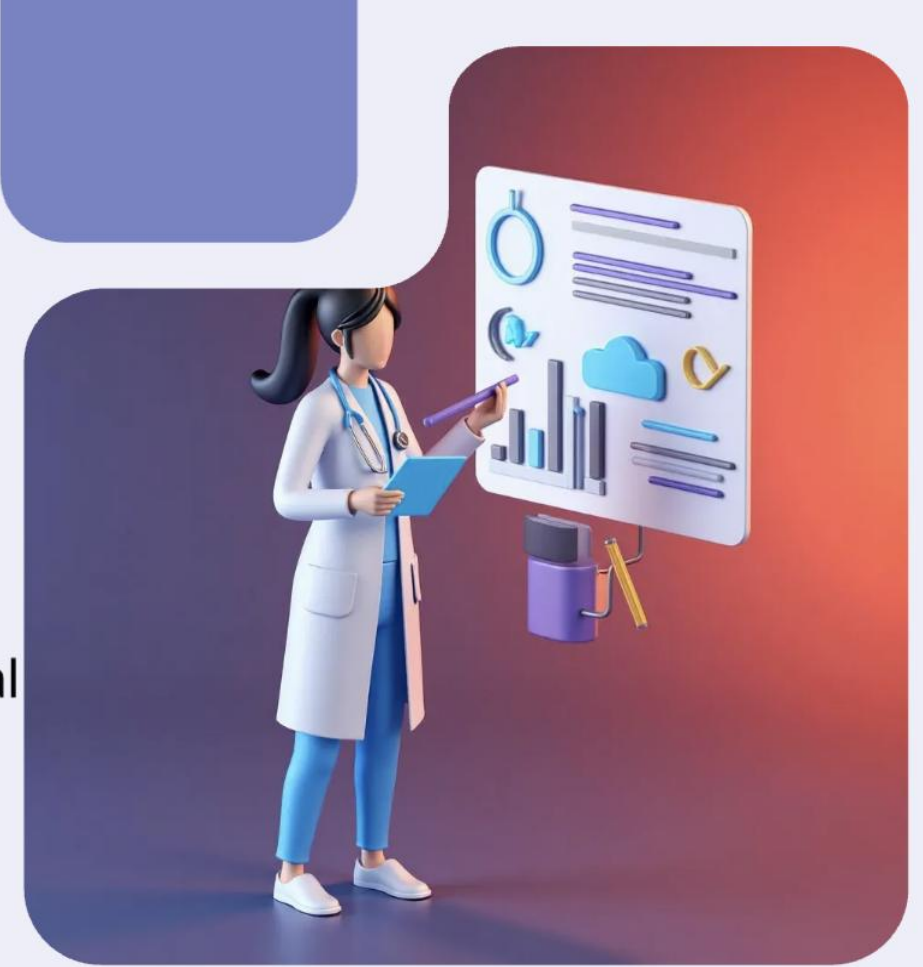
Dataset / Modality: Radiology and pathology images.

Key Findings: Balances compression speed and precision.

Strengths: Maintains strict lossless fidelity.

Limitations: May be slower on large datasets.

Relevance: Ideal for storage in medical archives and cloud platforms



Just Noticeable Difference (JND) Guided Wavelet Compression (2023)

Goal: Achieve clinically accurate compression that is lossless for diagnosis.

Methodology: Wavelet coding plus JND (Just Noticeable Difference). General medical pictures (radiology, MRI, CT) comprise the dataset and modality.

Key Findings: Preserves diagnostic features while eliminating perceptually unnecessary material.

Strengths: Maintains diagnostic quality while saving storage.

Limitations: Only "diagnostically" lossless, not bit-perfect.

Relevance: Useful for telemedicine storage and hospital PACS



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Methodology: DWT-based lossless compression scheme.

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Ref paper	Year	Method	Dataset/Modality	Compression Type	Achievements	Limitations
[1]	2025	Wavelet + Noise Robust	Ultrasound	Lossless / Near lossless	High compression, noise robust	Limited to ultrasound
[2]	2025	2D-DWT + EZW	General Medical	Lossy-to-Lossless	Efficient, low resource use	Small dataset
[3]	2024	Wavelet family comparison	General	Near lossless	Identified optimal wavelets	Not strictly lossless
[4]	2019	Wavelet + VQ	MRI/CT	Near lossless	Robust compression	Complexity, minor distortion
[5]	2023	Wavelet Difference Reduction	MRI/CT	Lossy/Near lossless	High diagnostic quality	Not pure lossless
[6]	2025	Deep Learning-based ...	Volumetric (CT/MRI)	Lossless	High bit-depth, efficient	High training cost
[7]	2017	3D Predictors + Entropy Coding	CT/MRI	Lossless	Fully lossless, high ratio	Computationally heavy
[8]	2005	Motion-Compensat...	4D MRI	Lossy-to-Lossless	Good for time-series	Outdated, limited scope
[9]	2022	DWT Lossless	Pathology, CT	Lossless	High precision, balanced speed	Limited large-scale test
[10]	2023	Wavelet + JND	Volumetric CT/MRI	Diagnostically Lossless	Maintains diagnostic features	Not strictly lossless

Module Preparation for Implementation



Dataset Collection & Preprocessing



Discrete Wavelet Transform (Compression)



Reconstruction (Decompression)



Performance Evaluation



Results & Visualization



Summary & Reporting

Module 1: Dataset Collection & Preprocessing



Objective: Gather medical image data and prepare it for compression

Tasks:



Select dataset (e.g., Brain Tumor Medical Image Dataset from Kaggle)



Convert images to grayscale for uniform processing



Resize images to a fixed resolution if necessary



Normalize pixel values (0–255)



Outcome: A clean, structured dataset ready for compression

Module 2: Discrete Wavelet Transform (DWT) Based Compression

Objective: Apply wavelet-based compression to medical images



Tasks:

- Perform 2D DWT decomposition on the medical image
- Store wavelet coefficients efficiently (using PyWavelets)
- Apply quantization or thresholding (if lossy compression is explored later)
- Encode coefficients in a compact representation



Outcome: A compressed representation of the medical image



Module 3: Image Reconstruction (Decompression)



Objective: Restore the original medical image from compressed data

Tasks:



Apply inverse DWT (IDWT) using stored coefficients



Reconstruct the image with minimal distortion



Outcome: Decompressed medical image visually close to original

Module 4: Performance Evaluation



Objective: Compare original vs. compressed images to ensure diagnostic quality

Tasks:



Compute MSE (Mean Squared Error) between original and reconstructed



Compute PSNR (Peak Signal-to-Noise Ratio)



Compute SSIM (Structural Similarity Index) for structural similarity



Generate error maps to visualize distortion



Outcome: Numerical and visual quality assessment of compression



Module 5: Results Visualization

Objective: Present compression results in a professional manner



Tasks:

- Display side-by-side images (original vs compressed)
- Show error heatmaps to highlight damaged regions
- Plot graphs for PSNR, SSIM across dataset images
- Summarize results in a performance table (DataFrame)



Outcome: Clear visualization of compression quality and reliability

Module 6: Summary & Reporting



Objective: Document and interpret findings

Tasks:



Summarize average compression performance (MSE, PSNR, SSIM)



Discuss trade-offs between compression ratio and image quality



Highlight importance of lossless compression in medical field (diagnostic accuracy)



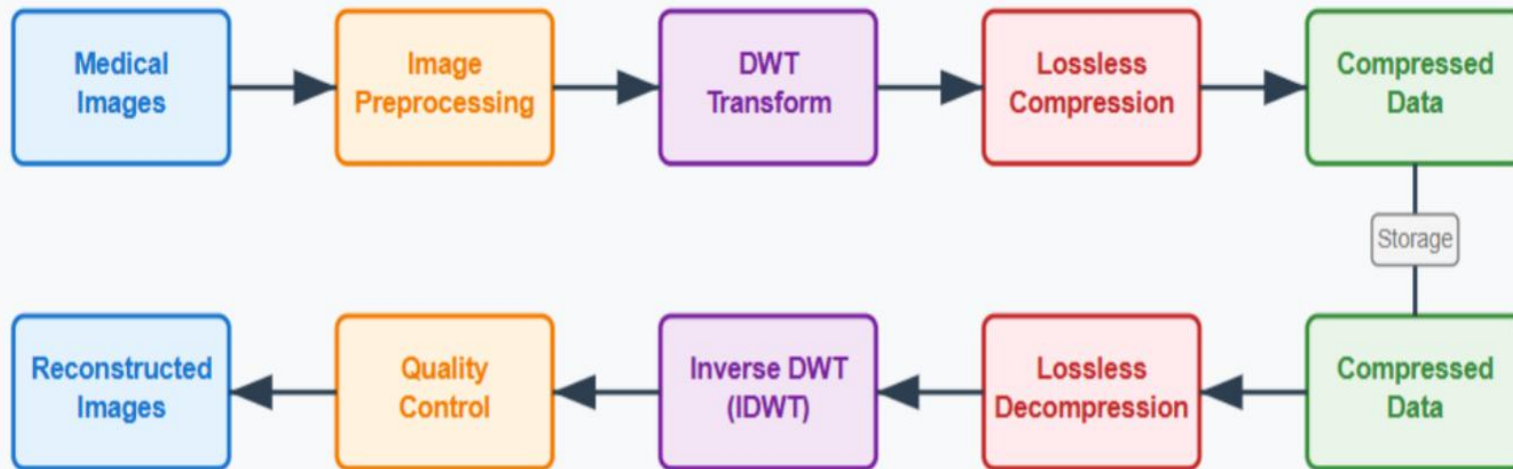
Prepare presentation-ready figures and tables



Outcome: A complete report module with results, ready for submission/presentation

Medical Image Compression Using DWT (Lossless)

Compression Path



Decompression Path

Key Features:

- Lossless compression maintaining diagnostic quality
- Support for DICOM, NIFTI, and other medical formats
- DWT-based transform for efficient compression
- Perfect reconstruction with quality verification



Downloadable link of dataset:

<https://www.kaggle.com/datasets/pkdarabi/medical-image-dataset-brain-tumor-detection>

Description of the Dataset An assortment of magnetic resonance imaging (MRI) scans of the human brain

An assortment of magnetic resonance imaging (MRI) scans of the human brain make up the "Medical Image Dataset: Brain Tumor Detection" dataset, which was donated by pkdarabi and obtained via Kaggle. Supporting research and experimentation in medical image analysis and brain tumor identification is the main goal of this collection. This dataset provides a reliable benchmark for researchers aiming to advance medical image processing, particularly in developing methods that ensure diagnostic quality preservation, which is critical in healthcare applications.

Key Highlights of dataset



Domain: Medical Imaging – Brain MRI Scans



Purpose: Designed for tasks such as brain tumor detection, classification, segmentation, and compression research
Content: High-quality MRI images, representing both tumor-affected and normal brain tissues



Applications:



Machine learning and deep learning model development



Image compression and reconstruction studies



Performance benchmarking of medical imaging algorithms



Computer-aided diagnosis (CAD) systems



Medical Image Compression Project Timeline

5-Week Development Cycle | DWT-Based Image Processing System



Week 1



Week 2



Week 3



Week 4



Week 5

PHASE 1

Literature Review & System Design

Research DWT techniques & medical imaging requirements

Week 1 | 7 Days

PHASE 2

Dataset Collection & Preprocessing

Gather medical images & implement preprocessing pipeline

Week 2 | 7 Days

PHASE 3

DWT Implementation & Testing

Develop compression & reconstruction algorithms

Week 3 | 7 Days

PHASE 4

Performance Evaluation & Optimization

Quality metrics analysis & system optimization

Week 4 | 7 Days

PHASE 5

Documentation & Presentation

Final report, visualization & project demo

Week 5 | 7 Days



Key Deliverables & Milestones

- System Requirements Document & Literature Survey
- Medical Image Dataset & Preprocessing Pipeline
- DWT Compression & IDWT Reconstruction Modules
- Performance Analysis & Quality Assessment Tools
- Complete Documentation & Project Presentation



Success Metrics & KPIs

- CR** Compression Ratio: >70% size reduction achieved
- PS** PSNR Quality: >40dB for medical image integrity
- SS** SSIM Index: >0.95 structural similarity maintained
- RT** Processing Time: <2 seconds per image
- DC** Documentation: Complete technical & user manuals

Month 1

Month 2

reference



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[2] Prayla Shyry, S., Prabu Shankar, K. C., & Hema, M. (2025). Efficient Medical Image Compression with 2D-DWT-EZW for Resource Optimization. International Journal of Uncertainty, Fuzziness & Knowledge-Based Systems, 33(5), 527–535. <https://doi-org.egateway.vit.ac.in/10.1142/S021848852540001X>



[3] Viswanathan, P., & Palanisamy, K. (2024). An Empirical Selection of Wavelet for Near-lossless Medical Image Compression. CURRENT MEDICAL IMAGING, 20, e300323215226. <https://doi-org.egateway.vit.ac.in/10.2174/1573405620666230330113833>



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[5] Zerva, M. C. H., Christou, V., Giannakeas, N., Tzallas, A. T., & Kondi, L. P. (2023). An Improved Medical Image Compression Method Based on Wavelet Difference Reduction. IEEE Access, Access, IEEE, 11, 18026–18037. <https://doi-org.egateway.vit.ac.in/10.1109/ACCESS.2023.3246948>



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